Chapter 12 Using Communications

This chapter describes how to use communications facilities, once you have set them up and connected to the other device, as described in Chapter 11.

12.1 Introduction

Hunter 16's communications ports can be used to connect to a very wide range of devices. These fall into two main categories: other computers and computer peripherals.

Communications can be used to exchange programs and data with other systems. For example:

- Programs can be loaded into **Hunter 16** daily, or as required, from central database. Use of optional secure protocols, or program-based error checks can guarantee accurate copies.
- New, revised or corrected programs can be returned to the database for storage. Database files and command functions can be accessed automatically, or under manual control using a terminal emulation.
- Data can be collected and down-loaded to mainframe, minicomputer or PC storage whenever required. By using one of the optional protocols, data transmissions can be formatted exactly to resemble existing data terminal configurations, allowing **Hunter 16** to slot directly into existing software support structures.
- Database information (for instance, tables of names and addresses) can be sent to **Hunter 16** for presentation to operatives in the field. Collected data can be linked to previously loaded database information for instant verification and validation.

12.2 Selecting the communications package

As already described, **Hunter 16** provides two communications packages:

- Simple IBM-compatible communications
- Sophisticated Husky communications

When you reboot, IBM communications will be selected. This is compatible with any applications that you are likely to want to use on **Hunter 16**.

Why then should you use Husky communications? As already described (see section 11.2), it provides a number of advantages. If you want these advantages, use Husky communications. Select and configure it by using the UTIL utility as described earlier in this chapter, and simple utilities (such as COPY) and any MS-DOS applications that do not re-initialise communications will use it automatically.

If you find that an application re-initialises communications, no harm will have been done; that application will simply use IBM communications.

Applications written specifically for **Hunter 16** may also select, configure and use Husky communications automatically.

Remember that **Hunter 16's** communications format is software controlled, so if **Hunter 16** fails to communicate, always check that the port is configured correctly.

12.3 Starting communication

When you turn **Hunter 16** on, the serial interface is left turned off to conserve power. It turns on automatically when a user program attempts to send or receive data. **Hunter 16** waits approximately 2 seconds to allow the interface to settle down before beginning transmission or reception.

While the port is turning on it may send spurious characters to the remote system, which may cause problems. To avoid this you can turn the port on immediately before connecting to the remote system. For example, by typing the following in MS-DOS command mode:

```
COPY CON COM1 [RETURN]
```

The port remains turned on until you turn **Hunter 16** off, so further transmissions are not affected in this way. If you are not going to use communications for a while, turn off again to save battery power. (The application may turn off the port automatically, as described in Part 3.)

12.4 Using standard facilities

The serial ports can be used from standard applications, as COM1 and COM2. As LPT1 is also defined as COM1, printed output will also normally be routed to COM1.

As supplied, communications can also be used in MS-DOS commands, by using the appropriate device name in place of a filename.

The main workhorse for file transfer is the COPY command. For example:

```
COPY COM1/A INPUT.TXT [RET]
```

would store input received from COM1 in the file INPUT.TXT, until an end-of-file marker ([Ctrl][Z]) is received. Note the /A switch; this makes sure that [Ctrl][Z] is interpreted as end of file.

WARNING: There is a problem when using the MS-DOS COPY command like this. If the transmitting system sends data slower than the receiving system requires it, the COPY command will timeout, inserting a NUL (ASCII 0) character in the input for each such timeout. This is unlikely to happen if the transmitting system is simply copying data from a fast disk (such as the **Hunter 16** RAM disk), but may cause problems if copying data from a slow peripheral or processing data before sending it, If this is the case, try a slower Baud rate or edit the NULs from the received file.

```
COPY LISTING.TXT COM2/A [RETURN]
```

would transmit the file LISTING.TXT Via COM2, ending with a [Ctrl][Z] (so the other system can detect the end of the file). Again, note the /A switch, which ensures that [Ctrl][Z] is sent. If the destination does not need to detect the end of the file (for example, is a printer), you could omit this.

These commands may be quite sufficient for transferring ASCII (text) files, but are not suitable for transferring binary files. To do this, you must either use a communications application (see section 12.8), or convert the binary data into a text format (such as Intel Hex), transfer that, then convert it back to binary format (see section 12.9.4).

COM1 is configured as LPTI automatically, so any command that prints will usually direct its output to COM1. For example:

[Prtsc] - output screen contents to COM1

[Ctrl][P] - toggle screen output to COM1 too

`PRINT` *filename* `[RET]` -queue a file for output to COM1 in background. (The PRINT command is not installed as standard.)

Standard input and output can also be redirected to the ports in the usual way. For example: -

`DIR >COM2 [RET]` - catalogue current directory and send result to COM2

`SORT <COM1 >INDEX [RET]` - input text from COM1, sort it and store the result in the file INDEX. (The SORT command is not installed as standard.)

12.5 Using the COM utility

COM is a simple file transfer utility which makes it easy to exchange files with another computer via the **Hunter 16's** serial ports. It provides binary transfer and two types of ASCII transfer, for compatibility.

12.5.1 Setting up

Use the UTIL utility to configure the speed, word length, parity and stop bit settings to suit the other computer. For maximum reliability, select **Hunter 16** mode and set CTS to ON and RTS to HOLD.

12.5.2 Command options

The COM command takes the following form:

`COM` *[path] filename [switches]*

The switches are:

- /R Receive a file and store it as [path]filename.
- /S Send the file (path]filename.
- /A Only valid with /S, this specifies ASCII transfer with transmission ending at the first Control-Z (1AH) character. (This is the default mode if not specified.)

/Z Only valid with /S, this specifies ASCII transfer with transmission ending after the first Control-Z (1AH) character has been sent.
/B Only valid with /S, this specifies binary transfer. Transmission ends when the number of bytes specified by the file's directory entry have been sent.
/1 Use COM1.
/2 Use COM2.

If you do not supply any switches, /R/1 is assumed. If you use /S without any other. switches, /A is assumed.

12.5.3 Sending a file

If you know that a file might contain Control-Z characters, use the /B switch to specify binary transfer.

If you are sending a text file, you can still use /B, but should use /A or /Z if the file has been created by an application which uses Control-Z for end of file (such as EDLIN) and then pads the file to some other length (eg a whole number of blocks). To decide whether to use /Z or /A, read the description of the COPY command in the **Hunter 16** DOS Manual.

While COM is transmitting it will display `Transmitting`, and then `Complete` when it has finished. If you see any other messages, see section 12.10.1.

12.5.4 Receiving a file

When you use COM to receive a file, it will display `Waiting` and then `Loading` while receiving data. Press [Ctrl][C] at any time to terminate the transfer; COM will display Aborted and finish writing the received file to disk. If you see any other messages, see section 12.10.1.

12.5.5 Examples

To send the text file FILES.LST complete with Control-Z, via COM1:

```
COM FILES.LST/S/2 [RET]
```

To send the program file PROG2.EXE from the C:\UTILS directory, via COM2:

```
COM C:\UTILS\PROG2.EXE/2/S [RET]
```

To receive data from COM2 and store it in C:\BACKUP\PROG.COM:

```
COM C:\BACKUP\ PROG.COM/R/2 G [RET]
```

12.6 Using Hunter 16 applications

If you are using an application which has been written especially for **Hunter 16**, it should handle communications automatically: it should select and configure communications to suit the remote system, then turn the port on and off as required and trap any errors for you.

Use of communications should normally have no effect on execution of the application. In normal operation, the only effect that you might see is a slight slowing down, if a lot of data is being transferred.

Refer to the instructions provided with the application, or ask your technical support team for further information.

12.6.1 Writing communications programs

Applications can be written to use communications almost as easily as console input/output, allowing the communications package to handle the complexities of data rates, handshaking, protocols and error correction.

The information that you'll require to write programs to communications is in Part 3 of this guide and in the *MS-DOS Programmer's Manual*. If you are writing BASIC programs, you may also need to refer to the GWBASIC manual.

12.7 Using MS-DOS applications

The serial ports are configured as COM1 andCOM2, so any application that allows you to use these devices can use communications. COM1 is also configured as LPTI, the device normally used for printer output, so printing will normally be routed to COM1.

If you are using a standard MS-DOS application such as an editor, wordprocessor, spreadsheet or accounts package, it probably won't provide any specialised communications facilities. In this case, you will probably only be able to output data while using it, by using the application's 'printing' options. This will normally be output via COM1, as described earlier. While you could use this to print results on a suitable printer, there's nothing to stop you connecting COM1 to another computer system which requires the information.

By default, when you reboot, IBM communications will be installed. You can use Husky communications by selecting it before running the application (eg by using UTIL) and gain all its benefits.

If you need to input files, or output files in formats not supported by the application's print option, you can use the standard MS-DOS facilities described in section 12.4.

12.8 Using MS-DOS communications packages

A number of third party communications packages are available for the IBM PC and compatibles. These typically offer very high data rates via direct connections (cables) and convenient menu-driven interface and automatic operation, including error-handling.

Such packages should work with **Hunter 16**, but because they are written for standard IBM PC hardware, they will not use any of the special facilities provided by Husky communications. They also may not work reliably at speeds greater than 19,200 Baud.

If you are at all unsure whether a package will work with **Hunter 16**, ask your supplier.

12.9 Example file transfer

The most important facility of a portable computer is its ability to communicate with other computers and peripheral devices. Hunter 16 uses the industry-standard RS232 interface and supports the common XON-XOFF protocol and a wide range of baud rates and data formats. This allows it to communicate with a wide range of computers, including microcomputer systems such as the IBM-PC. Optional protocols and internal upgrades extend this range even further.

This section describes how to exchange binary and text files with another system that supports 8-bit communication.

If the other system does not support 8-bit communications, binary data must be converted to a 7-bit format (such as Intel Hex format) before transfer, and converted back to binary format after it has been received. (See section 12.9.4.)

To transfer binary data you must use a communications application, or convert it into Intel Hex format, transfer that, then convert that back to binary format on the destination system. (Intel Hex facilities are built into most other Husky computers. Conversion utilities are also available from a number of sources for IBM PC compatibles such as **Hunter 16**.)

12.9.1 Exchanging files with another Hunter 16

There are many commercially available software packages for communications between IBM personal computers. The following method uses the supplied COM utility.

12.9.1.1 Setting up

Set and connect the two systems as described in section 11.7.1.

12.9.1.2 Transferring files

To transfer files between **Hunter 16s**, proceed as follows:

On the transmitting Hunter 16 (ie the one which is sending files), select MS-DOS command mode and type:

`COM` *filename* `/S/A [RET]` to transfer a text file, OR

`COM` *filename* `/S/B [RET]` to transfer a binary file.

On the receiving **Hunter 16**, select MS-DOS command mode and type:

`COM` *filename* `/R [RET]`

When transfer has finished the sender's screen should display `completed`.

Press [Ctrl][C] on the receiving Hunter 16.

If error messages are displayed on either screen, see section 12.10.

NOTE: See the warning about timeouts in section 12.4.

12.9.2 Exchanging files with Husky Hunter/Hawk

The procedure for exchanging files with the **Husky Hunter** and **Husky Hawk** is exactly the same, with the exception of the port configuration.

12.9.2.1 Setting up

Set up and connect the two systems as described in section 11.7.2/11.7.3.

12.9.2.2 Loading files

To transfer binary data to **Hunter 16** you must either use a communications application (see section 12.8) or Intel Hex format (see section 12.9.4).

To transfer ASCII (text) files, proceed as follows:

On **Husky Hawk/Hunter**, select DEMOS command mode and type:

```
SEND filename A
```

NOTE: Do not press enter yet!

On **Hunter 16**, select MS-DOS command mode and type:

`COM` *filename* `/S/A` [RETURN]

Now type [RETURN] on **Husky Hawk/Hunter**. You must type this within 20 seconds of finishing the command on **Hunter 16**.

You should see a series of messages on the **Husky Hawk/Hunter** screen, looking like this:

```
Transmitting
Waiting
Transmitting...
```

This indicates that transfer has begun. When the following message is displayed:

```
Completed
```

transfer has finished. **Hunter 16** should then also display

```
Completed.
```

If error messages are displayed on the **Hunter 16** Screen, see section 12.10. If any other error messages are displayed on the **Husky Hawk/Hunter** screen, see the appropriate User's Guide.

Note: See the warning about timeouts in section 12.4.

12.9.2.3 Sending files

You can send binary or ASCII files from **Hunter 16** as follows.

On **Husky Hawk/Hunter**, select DEMOS command mode and type:

```
INP 100 filename
```

NOTE: Do not press yet!

On **Hunter 16**, select MS-DOS command mode and type:

`COM` *filename* /S/A [RETURN] to send a text file, or

`COM` *filename* /S/B [RETURN] to send a binary file.

(Do not use the /Z switch as you do not need to send [Ctrl][Z])

Now press [RETURN] on Husky Hawk/Hunter.

Transfer should begin, as indicated by a series of messages on the **Husky Hawk/Hunter** screen, like this:

```
Waiting
Loading
Waiting
Loading...
```

When **Hunter 16** has finished sending, it will display Completed.

Check the **Husky Hawk/Hunter** display. It should be displaying the message:

```
Waiting
```

If not, wait for this message to appear. Press (Shift Esc) on the Husky Hawk/Hunter keyboard and the file will be stored on Husky Hawk/Hunter's disk. The following message will be displayed, but you may ignore it:

```
Aborted
```

If any other error messages are displayed on the **Husky Hawk/Hunter** screen, see the appropriate User's Guide. If error messages are displayed on the **Hunter 16** screen, see section 12.10.

12.9.3 Exchanging files with Husky Hawk 8/16

12.9.3.1 Setting up

Set and connect the two systems as described in section 11.7.3.

12.9.3.2 Loading files

To transfer binary data to Hunter 16 you must either use a communications application (see section 12.8) or Intel Hex format (see section 12.9.4).

To transfer ASCII (text) files, proceed as follows.

On **Husky Hawk 8/16**, select MS-DOS command mode and type:

```
ACOM [RETURN]
```

Type:

```
[T]
```

to select the Transmit option, select the Baud rate which you've

configured **Hunter 16** for, then type:

```
filename
```

NOTE: Do not press e yet!

On **Hunter 16**, select MS-DOS command mode and type:

`COPY COM1/A` *filename*

Now type [RETURN] on **Husky Hawk 8/16**. You must type this within 20 seconds of finishing the command on **Hunter 16**.

Transfer should begin. When the following message is displayed:

```
Completed
```

transfer has finished. **Hunter 16** should then display the following message:

```
1 File(s) copied
C>
```

If error messages are displayed on the **Hunter 16** screen, see section

12.10. If error messages are displayed on the **Husky Hawk 8/16** screen, consult the **Husky Hawk 8/16** User's Guide.

Note: See the warning about timeouts in section 12.4.

12.9.3.3 Sending files

You can send binary or ASCII (text) files from **Hunter 16** as follows.

On **Hunter 16**, select MS-DOS command mode and type:

```
COPY filename COM1
```

NOTE: Do not press [RETURN] yet!

On **Husky Hawk 8/16**, select MS-DOS command mode and type:

```
ACOM [RETURN]
```

Type:

```
[R]
```

to select the Receive option, select the Baud rate which you've configured **Hunter 16** for, then type:

```
filename
```

Now type on **Hunter 16**. You must type this within 20 seconds of finishing the command on **Husky Hawk 8/16**.

Transfer should begin. When the following message is displayed on Hunter 16:

```
1 File(s) copied
C>
```

transfer has finished. **Husky Hawk 8/16** should timeout in a few seconds, displaying the message:

```
Timeout
```

Press D and **Husky Hawk 8/16** should save the file, then return to MS-DOS command mode.

If error messages are displayed on the **Hunter 16** screen, see section 12.10. If error messages are displayed on the **Husky Hawk 8/16** screen, consult the **Husky Hawk 8/16** User's Guide.

12.9.4 Using Intel Hex format

If you cannot use 8-bit data transfer but need to transfer binary data, you must convert the files to a 7-bit format before transfer, send them, then convert them back to 8-bit on the destination system.

The commonest 7-bit format for binary files is Intel Hex format. Utilities to convert between Intel Hex format and binary are available for most computer systems, with names such as UNLOAD.

To convert files on **Hunter 16**, use any of the wide range of conversion utilities available for IBM PC compatibles. Conversion facilities already built into many other Husky computers, such as the **Husky Hunter** and **Husky Hawk**.

This process is obviously rather more laborious than straight 8-bit transfer. The transfer itself will also use more time as an Intel Hex format versions of a binary file is approximately twice as long. Intel Hex does however include facilities for automatic error-checking.

Use 8-bit transfer if you can.

12.10 Troubleshooting

If you cannot get communications to start:

- Check that the port is configured correctly to suit the cable and host.
- Check the cable and any intervening devices (such as modems, telephone lines, optoisolators or line buffers).
- Check that the host is ready to communicate.
- If possible, swap components (modems, cables, telephone lines) to help trace the problem.

If communications works but data is corrupted:

- Check that the data formats and speeds are the same for Hunter 16 and the host.
- Check that all connections are noise-free.
- Try a slower speed or more secure protocol if one is available.

If communications fails after operating correctly for a time:

- Check for noisy lines (or use a slower speed or more secure protocol).
- Check that the connection is still made (eg check that the telephone line has not been lost or a cable has not come unplugged).
- Check for component failure.

If communications fails because of timeouts, you can increase the periods or disable the timeouts entirely by using the UTIL command or by using similar options in a **Hunter 16** application.

12.10.1 Error messages

The following error messages may be displayed by the COM utility:

`Unable to open file`
You have specified a file to send which does not exist.

`Unable to create file`
You have specified a file to receive which is not a valid DOS file name, or there is already a file with this name which is write-protected.

`Invalid file name`
You have not specified a file name, or the file name is not a valid DOS file name.

`Error writing to file`
COM could not write received data to the disk. The most likely reason is a lack of disk space.

`File read error`
COM could not read the some or all of the information from the file that you specified to send. (Use CHKDSK to check the file.)

`Communications error`
Either the communications settings on the two computers do not match, or the cable is faulty.

`Receive options not allowed`
You have typed the /A, /B or /Z option with /R. These options can only be used with /S.

`Conflicting or too many options`
You have typed two conflicting options, such as /S and /R or /1 and /2.

`Unknown option`

You have typed an option which is not one of /1, /2, /A, /B, /R, /S and /Z.